THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 13: **Two Variables, One Line** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The line $17x + 18y = 306$ cuts the axes at two points. The area of the triangle it forms with the axes (square units) is: 1

(a) 155 **(b)** 153 **(c)** 152 **(d)** 154

2. A taxi fare is ₹243 for 5 km and ₹425 for 12 km (fare is a linear function of distance). The fare for 38 km (in ₹) is: 1

(a) 1127 **(b)** 1214 **(c)** 1101 **(d)** 1075

3. The line $4x + y = 9$ is parallel to the line $kx + 5y = 11$. The value of k is: 1

(a) 21 **(b)** 20 **(c)** 22 **(d)** 19

4. The points $(-5, 19)$, $(-4, 18)$ and $(-3, k)$ are collinear. The value of k is: 1

(a) 18 **(b)** 19 **(c)** 17 **(d)** 16

5. The pair $kx + 3y = k - 3$ and $12x + ky = k$ has infinitely many solutions. The value of k is: 1

(a) 6 **(b)** 7 **(c)** 9 **(d)** 3

6. The line $9x + 5y = 45$ has intercepts p on the x-axis and q on the y-axis. The value of $p + q$ is: 1

(a) 14 **(b)** 15 **(c)** 59 **(d)** 13

7. Using $F = \frac{9}{5}C + 32$, the temperature $90°$C in degrees Fahrenheit is: 1

(a) 194 **(b)** 195 **(c)** 193 **(d)** 203

8. The slope of the line through $(12, 22)$ and $(-24, 23)$ is: 1

(a) $\frac{71}{36}$ **(b)** $-\frac{1}{36}$ **(c)** $\frac{35}{36}$ **(d)** $-\frac{37}{36}$

9. The line $24x + 11y = 264$ cuts the axes at two points. The area of the triangle it forms with the axes (square units) is: 1

(a) 133 **(b)** 132 **(c)** 131 **(d)** 134

10. A taxi fare is ₹300 for 4 km and ₹600 for 10 km (fare is a linear function of distance). The fare for 29 km (in ₹) is: 1

(a) 1500 **(b)** 1650 **(c)** 1550 **(d)** 1600

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. For what value of k does the pair $kx + 4y = k - 4$ and $16x + ky = k$ have infinitely many solutions? 2

12. Find the area of the triangle formed by the line $9x + 15y = 135$ and the coordinate axes. 2

13. Find the equation of the line through $(-12, -24)$ and $(-9, -21)$ in the form $y = mx + c$. 2

14. For what value of k does the pair $kx + 6y = k - 6$ and $24x + ky = k$ have infinitely many solutions? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A line cuts the x-axis at $(18, 0)$ and the y-axis at $(0, 17)$. Find its equation, its slope and the area of the triangle it makes with the axes. 3

16. A water supplier charges a fixed amount plus a rate per kilolitre. For 4 kL the bill is ₹107 and for 11 kL it is ₹212. Find the equation of the bill, and the usage whose bill is ₹317. Also find the bill for no usage. 3

17. Find k if the points $(19, -9)$, $(21, -13)$ and $(23, k)$ lie on one straight line. 3

OR

A line cuts the x-axis at $(15, 0)$ and the y-axis at $(0, 18)$. Find its equation, its slope and the area of the triangle it makes with the axes. 3

18. A water supplier charges a fixed amount plus a rate per kilolitre. For 4 kL the bill is ₹116 and for 10 kL it is ₹212. Find the equation of the bill, and the usage whose bill is ₹324. Also find the bill for no usage. 3

OR

Find k if the points $(-18, -17)$, $(-17, -15)$ and $(-16, k)$ lie on one straight line. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A line passes through $(-1, 1)$ and $(5, 7)$. Find its equation, then find k if the point $(k, 9)$ lies on it, and the x-intercept. 5

OR

A taxi fare is ₹176 for 2 km and ₹476 for 14 km. Assuming the fare is linear in the distance, find the fare per km, the fixed charge and the fare for 27 km. 5

20. A line passes through $(0, 2)$ and $(1, 6)$. Find its equation, then find k if the point $(k, 30)$ lies on it, and the x-intercept. 5

OR

A taxi fare is ₹222 for 4 km and ₹310 for 8 km. Assuming the fare is linear in the distance, find the fare per km, the fixed charge and the fare for 38 km. 5

END OF QUESTION PAPER — Total: 40 Marks